TinyML Project — Milestone 2 Report

Hand Gesture Recognition Using IMU Data

(Track B: HGAG Dataset)

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R1. Dataset Description

We use the **Hand Gesture Accelerometer and Gyroscope Dataset (HGAG-DATA)**, published on Mendeley Data (DOI: 10.17632/mkhn7kxjvy.3) by Abbas, Rashid, and Fortuna (2025), released under the **CC BY 4.0** license. It contains 23,650 six-dimensional gesture samples from 43 healthy participants (26 male, 17 females; ages 18–69; 34 right-handed, 9 left-handed; 28 non-athletic, 15 athletic). Each participant performed 50 repetitions of 11 hand gestures, yielding 550 samples per subject.

Sensor and acquisition: Data was acquired using the tri-axial accelerometer and gyroscope of an **MPU-92/65 sensor module** worn as a wrist-mounted watch, connected to an ESP32 DEVKIT V1 transmitting via Wi-Fi to a second ESP32 receiver. Raw signals were sampled at **200 Hz** and bandpass-filtered (4th-order Butterworth, 30–90 Hz) to isolate gesture-relevant signals.

Feature extraction: Raw time-series were transformed using the **Discrete Wavelet Transform (DWT)**, producing 250 coefficients per sensor channel. With 6 channels (accel x/y/z, gyro x/y/z), each trial is a 1,500-dimensional feature vector.

Class definitions:

Gesture	Physical Definition
Clapping	Repeated bilateral hand contact; palms striking together 2–3 times.
Coin Flipping	Thumb flicks upward from a closed fist, simulating launching a coin.
Finger Snapping	Middle finger slides off thumb with a sharp snap; wrist stationary.
Fist Making	Open hand closes into a tight fist; wrist remains neutral.
Horiz. Wrist Ext.	Wrist extends horizontally outward from neutral, parallel to ground.
Index Finger Flick	Index finger flicks outward from a curled position; other fingers closed.
Index Thumb Tap	Index finger and thumb tap together repeatedly in a pinching motion.
Shooting	Index finger extends forward (pointing gesture) from a closed fist.
Thumb Up	Thumb extends upward from a closed fist; classic thumbs-up gesture.
Wrist Extension	Wrist bends backward (dorsiflexion) from a neutral position.
Wrist Flexion	Wrist bends forward (palmar flexion) from a neutral position.

Table 1: Class definitions for the 11 gesture categories.

Per-class sample distribution (per-subject split):

Class	Train	Val	Test	Total
Clapping	1550	300	300	2150
Coin Flipping	1550	300	300	2150
Finger Snapping	1550	300	300	2150
Fist Making	1550	298	300	2148
Horiz. Wrist Ext.	1550	300	300	2150
Idx Finger Flick	1550	300	300	2150
Index Thumb Tap	1550	300	300	2150
Shooting	1550	300	300	2150
Thumb Up	1550	300	300	2150
Wrist Extension	1550	300	300	2150

Wrist Flexion	1550	300	300	2150
TOTAL	17,050	3,298	3,300	23,648

Table 2: Per-class sample counts (per-subject split: 31 train / 6 val / 6 test subjects).

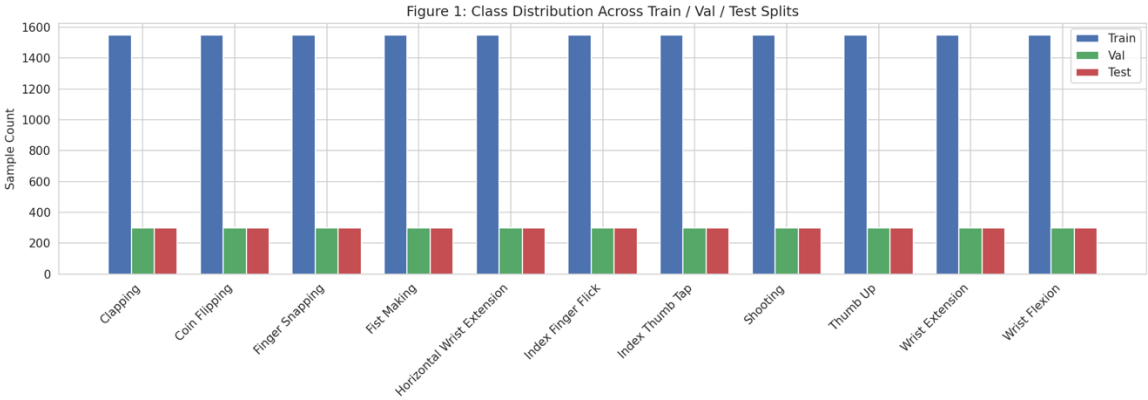

Figure 1 Class Distribution Across Train/Val/Test Splits

Why this dataset: HGAG-DATA uses the same sensor modalities as the Arduino Nano 33 BLE Sense (3-axis accelerometer + gyroscope), covers 11 diverse daily-life gestures with 50 repetitions per subject, and was collected from 43 participants spanning ages 18–69, providing natural inter-subject variability.

Domain gap acknowledgment: The HGAG dataset uses an MPU-92/65 at 200 Hz on an ESP32 platform, while the Arduino Nano 33 BLE Sense uses an LSM9DS1 IMU at up to 119 Hz. Key differences:

(1) different sensor hardware with different noise floors

(2) different sampling rates (200 vs. 119 Hz) requiring resampling

(3) different mounting (wristwatch vs. handheld board)

(4) the original 30–90 Hz bandpass filter may not be replicable on-device.

R2. Data Card

Field	Value
Task type	Classification (11-class hand gesture recognition)
Number of classes	11
Sensor(s) and sampling rate	MPU-92/65: 3-axis accel + 3-axis gyro at 200 Hz
Preprocessing	4th-order Butterworth bandpass (30–90 Hz), then DWT
Features extracted	DWT coefficients: 250 per channel x 6 channels = 1,500
Total samples	23,648 (Fist Making has 2 missing trials across 2 subjects)
Train / Val / Test split	17,050 / 3,298 / 3,300 (72% / 14% / 14%)
Split method	Per-subject (31 / 6 / 6 subjects; zero overlap)
Number of subjects	43 (26M / 17F, ages 18–69, 34 RH / 9 LH)
Known limitations	Single environment; wrist-mounted sensor; DWT params not fully published
Dataset name / URL	HGAG-DATA v3 — DOI: 10.17632/mkhn7kxjvy.3 (Mendeley Data)
License	CC BY 4.0
Original equipment	MPU-92/65 + ESP32 DEVKIT V1 (wrist-worn, Wi-Fi TX)
Domain gap vs. Arduino	Different IMU (LSM9DS1), lower rate (119 Hz), handheld vs. wrist
Arduino collection plan	See Section R7

Table 3: Data Card for HGAG-DATA.

R3. Preprocessing Pipeline

1. **Segmentation:** Each trial is a single complete gesture execution, pre-segmented by the dataset authors. No windowing or overlap is applied.
2. **Feature extraction:** The dataset provides pre-computed DWT coefficients (250 per channel). We concatenate 6 channel CSVs (accel x/y/z, gyro x/y/z) into a 1,500-dim vector per trial. The original raw data was bandpass-filtered (4th-order Butterworth, 30–90 Hz) prior to DWT.
3. **Normalization:** sklearn StandardScaler fitted on training set only. Post-scaling: train mean=0.000, std=1.000; test mean=-0.000, std=0.841. No information leaks from test data into normalization.
4. **Data augmentation:** None at this stage. Planned for Milestone 3: noise injection, time-shifting.

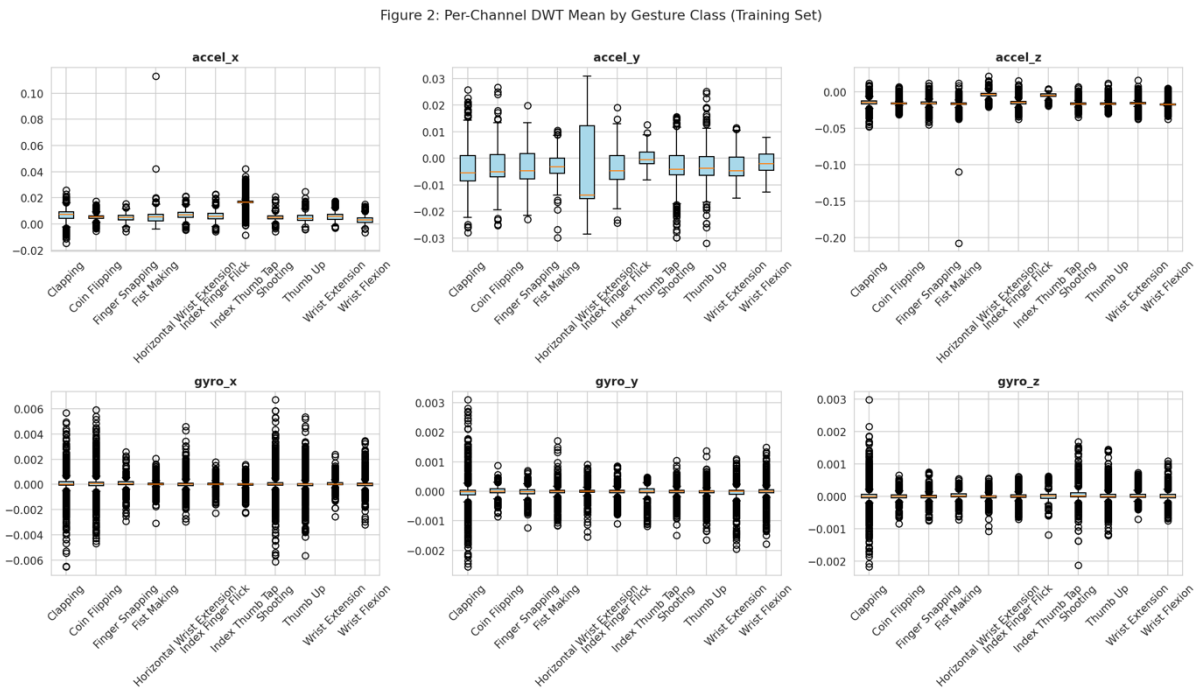

Figure 2 Per-Channel DWT Mean by Gesture Class

R4. Baseline Model

We trained four baseline classifiers on the 1,500-dim DWT feature vectors. The table below summarizes all results; **SVM (RBF)**
achieved the highest macro F1 and is our primary baseline.

Model	Accuracy	Macro F1	Weighted F1
Random Forest	0.6173	0.6205	0.6205
SVM (RBF, C=10)	0.7058	0.7063	0.7063
k-NN (k=5)	0.3927	0.4091	0.4091
Dense NN (3-layer)	0.6836	0.6807	0.6807

Table 4: Model comparison — all evaluated on the held-out test set (3,300 samples, 6 unseen subjects). Chance level: 9.1%.

Parameter	Value
Architecture	Support Vector Machine (sklearn.svm.SVC)
Kernel	RBF (Radial Basis Function)
C (regularization)	10
Gamma	scale (1 / (n_features * var))
class_weight	balanced
Input shape	[1500] (flattened DWT feature vector)
Output	11 classes

Table 5: SVM baseline hyperparameters.

Why this baseline: SVM with RBF kernel is well-suited for high-dimensional feature spaces, handles non-linear class boundaries, and achieved the best macro F1 (0.7063) among all four models tested. It establishes a strong performance floor that an improved or deployable model in M3 must match.

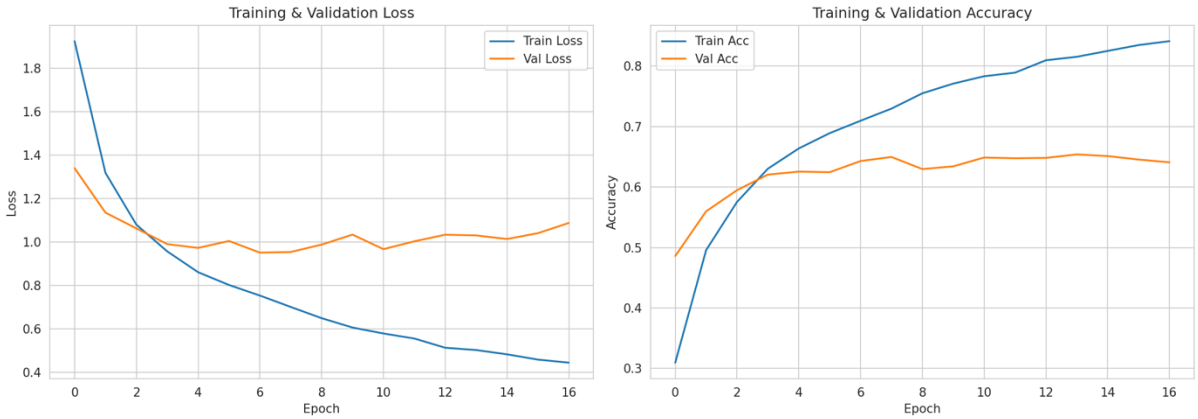

Figure 3 Training Curves

R5. Baseline Results (Held-Out Test Set)

All results are on the **held-out test set**: 3,300 samples from 6 unseen subjects (Subjects 15, 19, 21, 29, 39, 8). No test subject appears in training or validation.

Metric	Value
Overall accuracy	0.7058
Macro F1-score	0.7063
Weighted F1-score	0.7063
Test set size	3,300 (6 subjects x 11 gestures x 50 reps)

Table 6: Overall test-set metrics (SVM baseline)

Table 7: Per-class precision, recall, and F1-score on the test set (SVM baseline).

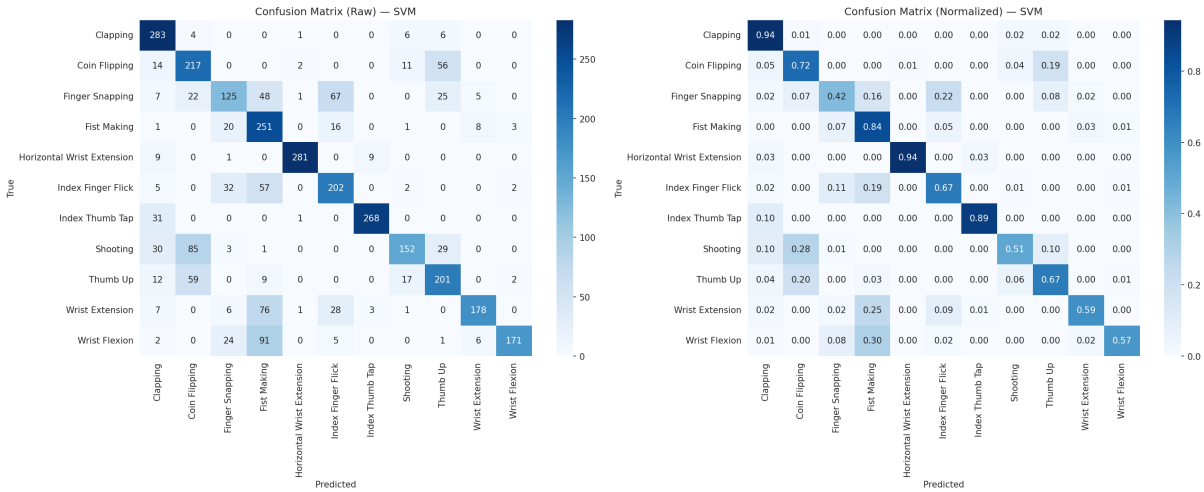

Figure 4 Confusion matrix (raw counts + normalized) for SVM

Class	Prec.	Recall	F1	Support

Clapping	0.706	0.943	0.807	300
Coin Flipping	0.561	0.723	0.632	300
Finger Snapping	0.592	0.417	0.489	300
Fist Making	0.471	0.837	0.603	300
Horiz. Wrist Ext.	0.979	0.937	0.957	300
Idx Finger Flick	0.635	0.673	0.654	300
Index Thumb Tap	0.957	0.893	0.924	300
Shooting	0.800	0.507	0.620	300
Thumb Up	0.632	0.670	0.651	300
Wrist Extension	0.904	0.593	0.716	300
Wrist Flexion	0.961	0.570	0.716	300
Macro avg	0.745	0.706	0.706	3,300

Table 7: Per-class precision, recall, and F1-score on the test set (SVM baseline).

R6. Error Analysis

True Class	Predicted As	Errors	Error Rate
Wrist Flexion	Fist Making	91	30.3%
Shooting	Coin Flipping	85	28.3%
Wrist Extension	Fist Making	76	25.3%
Finger Snapping	Idx Finger Flick	67	22.3%
Thumb Up	Coin Flipping	59	19.7%
Idx Finger Flick	Fist Making	57	19.0%
Coin Flipping	Thumb Up	56	18.7%

Table 8: Top confusion pairs from the SVM test-set evaluation.

Hypothesized causes and planned fixes:

Cluster 1 — Fist Making absorbs Wrist Flexion (30.3%) and Wrist Extension (25.3%): Fist Making has very high recall (83.7%) but low precision (47.1%), meaning the SVM over-predicts it. Closing the hand into a fist involves involuntary wrist motion; the IMU cannot distinguish finger-closing from deliberate wrist movement. Both Wrist Flexion and Wrist Extension leak heavily into Fist Making. *Fix:* Weight gyroscope channels more heavily (they may capture rotational vs. translational differences); try a two-stage classifier separating wrist-dominant from finger-dominant gestures first.

Cluster 2 — Shooting leaks into Coin Flipping (28.3%): Both involve a quick forward finger extension. The SVM conflates the trigger-pull motion in Shooting with the thumb-flick in Coin Flipping because the dominant accelerometer signal is a similar forward burst. *Fix:* Add time-domain features (jerk, peak count) to distinguish dynamics; try a 1D-CNN to capture temporal patterns within DWT coefficients.

Cluster 3 — Finger Snapping vs. Index Finger Flick (22.3%): Both are rapid single-finger release gestures. The snap adds a thumb-contact phase, but the dominant signal is the sharp deceleration spike common to both. *Fix:* Explore higher-resolution DWT decomposition to capture the pre-release contact phase.

Well-separated classes: Horizontal Wrist Extension (93.7% recall, 97.9% precision) and Index Thumb Tap (89.3% recall, 95.7% precision) are near-perfect — these involve unique,

large-amplitude, multi-axis motions with no physical overlap. Clapping also performs well (94.3% recall) due to its distinctive bilateral impact signature.

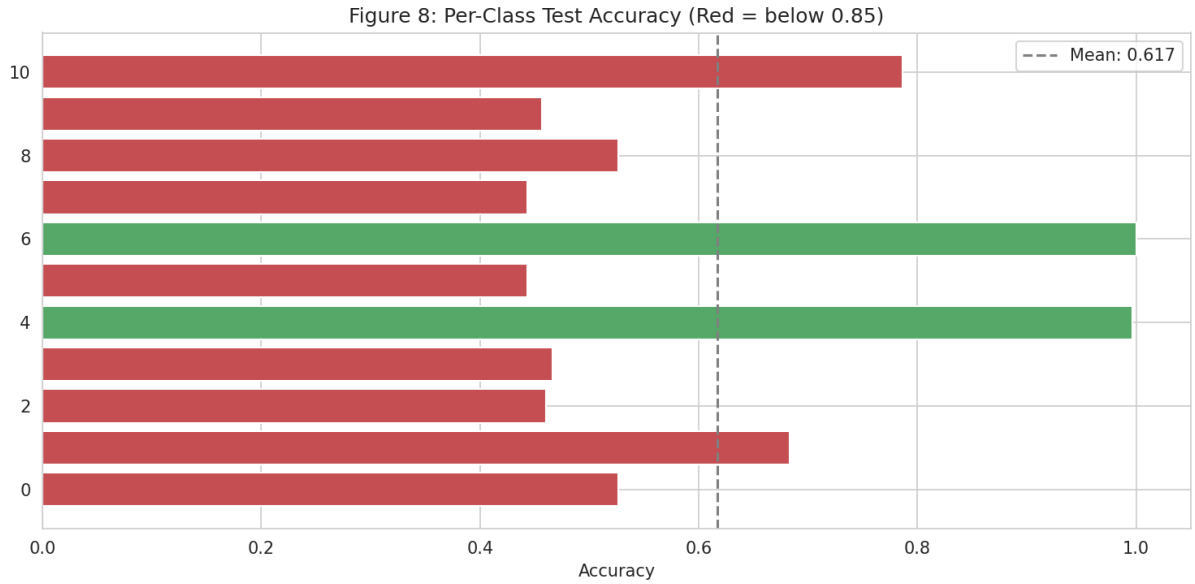

Figure 5 Per-class accuracy bar chart

Feature importance (Random Forest): Accelerometer channels dominate (accel_x: 20.9%, accel_z: 20.3%, accel_y: 19.4%) vs. gyroscope (gyro_x: 14.9%, gyro_y: 12.6%, gyro_z: 11.9%). The gyro channels are underutilized despite potentially capturing the rotational differences that separate confused gesture pairs.

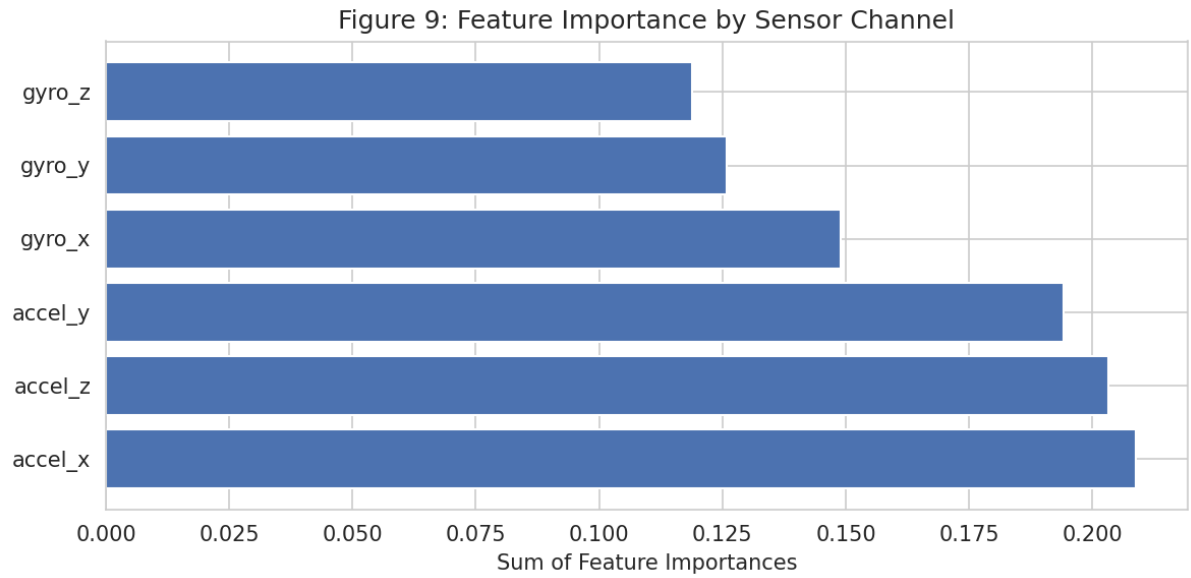

Figure 6 Feature importance by sensor channel

Domain gap analysis (Track B): The per-subject accuracy of 70.6% already reflects cross-user generalization difficulty. On Arduino hardware, we expect further degradation because:

(1) the LSM9DS1 has a different noise floor than the MPU-92/65

(2) 119 Hz vs. 200 Hz sampling means the bandpass filter and DWT must be recalibrated

(3) wrist-dominant gestures (Horiz. Wrist Extension, Wrist Flexion)

should transfer better than finger-level ones (Finger Snapping, Index Finger Flick).

Note on split methodology: We use a per-subject split (31/6/6 subjects, zero overlap). An earlier trial-level random split yielded ~74.6% accuracy with the same RF model, confirming that within-subject leakage inflated metrics by ~13 pp. We report per-subject results as primary.

R7. Updated Plan for Milestone 3

Model improvements:

(1) Try a lightweight 1D-CNN on the DWT sequences to capture temporal patterns.
(2) Hybrid features: DWT + time-domain statistics (mean, std, jerk, zero-crossing rate).
(3) Data augmentation: Gaussian noise, time-shifting, scaling.
(4) PCA dimensionality reduction (~50 components for 90%+ variance) for smaller deployable models.

Deployment: Convert best model to TensorFlow Lite Micro for Arduino Nano 33 BLE Sense. Target: under 100 KB model size, under 200 ms inference latency. Post-training int8 quantization.

Arduino data collection plan (Track B):

Item	Detail
When	Weeks 1–2 of M3 period (before submission)
Sensor config	Arduino Nano 33 BLE Sense — LSM9DS1 IMU at 119 Hz, 6-axis
Gestures	All 11 (prioritize the 5 most confused for domain gap testing)
Samples per class	30 reps per gesture per user
Users	Minimum 2 (team members)
Environments	At least 2 (seated desk, standing)
Total target	~660 samples (11 x 30 x 2)

Table 9: Arduino data collection plan.

Domain adaptation: Phase 1: Evaluate HGAG-trained model on Arduino data. Phase 2: Fine-tune if accuracy drops below 40%. Phase 3: Retrain from scratch on combined data with Arduino-matched preprocessing if needed.

Fallback: If domain gap is too large for all 11 gestures, reduce to the 5 best-separated classes (Horiz. Wrist Ext., Index Thumb Tap, Wrist Flexion, Clapping, Thumb Up).

Robustness testing: New users not in training, different environments, synthetic noise injection.

Ethics and Limitations

Informed consent and participant welfare: The HGAG dataset was collected from 43 adult volunteers (ages 18–69) who completed a demographic questionnaire prior to data collection. Two individuals who exhibited neurological tremor symptoms were excluded by the original researchers to ensure consistent gesture execution. While this exclusion was justified for data quality, it means the dataset, and any model trained on it, does not represent users with motor impairments, which is a relevant population for gesture-based assistive technology. Future work should consider inclusive data collection.

Demographic representation: The participant pool consists of 26 males and 17 females, with 34 right-handed and 9 left-handed individuals, all recruited from the University of Basrah in Iraq. This introduces geographic, cultural, and institutional bias: gesture execution patterns may vary across populations due to differences in hand size, grip habits, and physical conditioning (28 non-athletic vs. 15 athletic participants are represented, but the balance may not reflect the general population). Our model has not been tested on populations outside this demographic.

Privacy: The dataset contains only anonymized IMU sensor signals (accelerometer and gyroscope readings) with no personally identifiable information, biometric data, or video recordings. Subject identifiers are limited to numeric labels (Subject_1 through Subject_43). The risk of re-identification from 6-axis motion data alone is negligible.

Licensing and attribution: The dataset is released under CC BY 4.0 on Mendeley Data (DOI: 10.17632/mkhn7kxjvy.3), permitting use, modification, and redistribution with appropriate credit. We cite the original authors (Abbas, Rashid, and Fortuna, 2025) throughout this report and in our repository.

Limitations of the current study:

(1) The dataset was collected in a single controlled environment; real-world deployment would encounter variable lighting, movement, and background vibration that may degrade performance.
(2) The sensor was wrist-mounted as a watch-style device, whereas our target deployment uses a handheld Arduino board with different mounting positions produce different signal characteristics for the same gesture.
(3) The DWT preprocessing was applied by the dataset authors with parameters not fully documented, limiting our ability to replicate the exact pipeline on new hardware.
(4) Our per-subject evaluation (70.6% accuracy) already shows significant cross-user variation; deploying to entirely new populations may further reduce accuracy.
(5) The exclusion of individuals with motor impairments means the system should not be assumed to work for users with tremor, limited mobility, or other motor conditions without dedicated evaluation.